

Luis Martinez

Electrical Engineer

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PROFESSIONAL EXPERIENCE

SPACE INFORMATION LABORATORIES

Electrical Engineer II

2025-Present

- Test System Design
 - Helped design electrical test fixtures and procedures for PCBA DVT
 - Designed system test equipment PCBAs to facilitate streamlined testing
- R&D Proposal Support
 - Assisted with proposal efforts by conducting trade studies and generating preliminary ICDs
- Circuit Analysis
 - Used worst-case circuit analysis (WCCA) to evaluate BMS circuit compliance to PMAM Rev B and RCC-319
 - Failure Mode, Effects, and Criticality Analysis (FMECA) for Autonomous Flight Termination Unit (AFTU) and BMS

RAYTHEON INTELLIGENCE AND SPACE

Electrical Engineer II

2024-2025

Electrical Engineer I

2022-2024

- PCB Design
 - Designed two boards based on customer requirements which were brought through CDR.
 - Used Hyperlynx SI/PI to run power and signal integrity of routed PCBs
 - Created PWB, PWA, parts list, and schematic drawings
- Circuit Analysis
 - Used worst-case circuit analysis techniques (WCCA) to evaluate the performance of various power electronics circuits
- Firmware Design
 - Implemented and debugged raw HDL for Intel SoCs in VHDL using Signaltap, Quartus Pro, and Qsys
 - Used Simulink and Matlab for higher level system design
- Lab Experience
 - Conducted system level environmental tests with vibe/temperature chambers
 - Frequent work with signal generators, DC power supplies, spectrum analyzers,, and oscilloscopes

BOSTON SCIENTIFIC

Electrical Engineering Intern

2021

- Designed a PCB to model behavior of the spine under certain stimuli
- Wrote C for a microcontroller on the PCB to define response to input data
- Used concepts of DSP, digital and analog filtering, and digital to analog conversion

PERSONAL PROJECTS

Guitar Headphone Amplifier

- Designed a board in KiCad to amplify an electrical guitar signal to drive a pair of headphones

IMU Attitude Estimation with Quaternions

- Calculated and visualized heading using an IMU, Arduino Nano Every, and quaternions

SKILLS

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|--------------------------------|-------------------------------|
| • Xpedition Enterprise | • FMECA |
| • Circuit Simulation (LTSpice) | • Python |
| • VHDL, Verilog, SystemVerilog | • Altium |
| • C/C++ | • Embedded Systems |
| • Matlab/Simulink | • VPX/VITA Standard Knowledge |

EDUCATION

CORNELL UNIVERSITY

Ithaca, NY

Bachelor of Science in Electrical and Computer Engineering

2018-2022

- Coursework centered around Embedded Systems, Digital Signal Processing, and FPGA development
- Electrical lead for Cornell Nexus which strived to create a robot that filters microplastics from beaches. Worked closely with motors and motor controllers, motor encoders, Raspberry Pi's, and batteries.